Recognising without recalling:

a small recurrent memory recognises long after it stops recalling

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benchmark: `irodkin/recognition-memory`

Takeaway

A memory can tell whether an input is familiar long after it can no longer reproduce it.

- Shrinking a GDN-2 state, **exact recall fails first**: at 16×8, $N\geq128$ no sequence is copied exactly, yet surprise still scores acc9 ≥0.98.
- At the smallest state (16×4, $N=256$) the memory **trained for recognition** wins: acc9 **0.81** vs. **0.43** for surprise (chance 0.11), with 18% of tokens recalled.
- Training that classifier at long N needs a **length curriculum**.

Hypothesis

Memory need not *reproduce* the past. It can instead *recognise* whether an input is consistent with it, using low **surprise** as the familiarity signal. Checking is easier than generating, so recognition should need **less capacity** than recall.

Benchmark: one sequence, eight near-copies

Uniform i.i.d. tokens, vocabulary 16 (4 bits each), $N = 4 \dots 256$. 900k / 50k / 100k sequences. Each comes with **8 corrupted copies**: one forced change plus $p = \frac{1}{N-1}$ elsewhere, so **2 changes on average**.

true	3	a	0	f	7	7	c	1	9	e	4	2	b	5	d	8
copy	3	a	0	f	7	2	c	1	9	e	4	2	b	5	d	8

Task. Score the true sequence and its 8 copies.
acc9: true one ranked first (chance $1/9$). **pairwise**: true beats one copy (chance $1/2$).

Capacity, in theory. Recall needs all $4N$ bits (1,024 at $N=256$). Recognition needs a few: keeping only the token sum mod 16 (4 bits) gives acc9 0.85; adding a position-weighted sum (8 bits) gives 0.965, at *every* N.

Model: one network, two readouts

2-layer Gated DeltaNet-2 (`fla`), width 128, one head. The only memory is the recurrent state, $d_k\times d_v$ per layer: 128×128 (16,384 numbers) down to 16×4 (64).

state $S_t \in \mathbb{R}^{d_k\times d_v}$ carries x forward

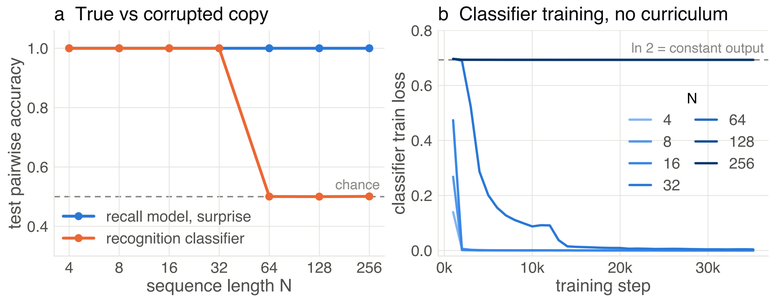
BOS | **sequence** $x_{1..N}$ | **SEP** | **candidate** $c_{1..N}$

Recall model
trained to copy x (next-token CE)
score $=$ −surprise
$= \sum_j \log p(c_j \mid x, c_{<j})$

Recognition model
trained on "is $c = x$?"
(BCE, one label per sample)
score $=$ logit at the last token

1 Full-size state: recall is easy, the classifier never starts

At 128×128 the **recall model** copies every length almost perfectly (exact copy 0.988 at $N=256$), and its surprise ranks the true sequence first in *every* test sample. The **recognition classifier** fails from $N=64$ on.

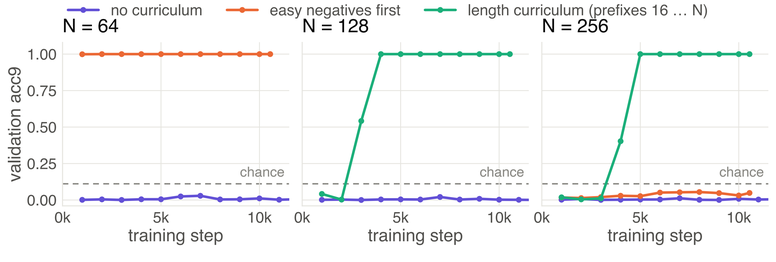


Test on 100k sequences, 10 epochs, seed 0. (b) Classifier BCE, mean over 1,000 steps, one line per N.

- The loss sits at $\ln 2$ for all 35k steps: a constant output, gradients decay to ≈0.03.
- Same network learns recall in $<1{,}000$ steps ⇒ an **optimisation** failure, not a capacity limit. One label per sample says little about 2 differing tokens out of 256.

2 A length curriculum lets the classifier learn

For the first half of training, train on prefixes $16, 32, \dots, N$ (equal time each), with easy, heavily corrupted negatives fading out. Then switch to full length and the benchmark's copies.

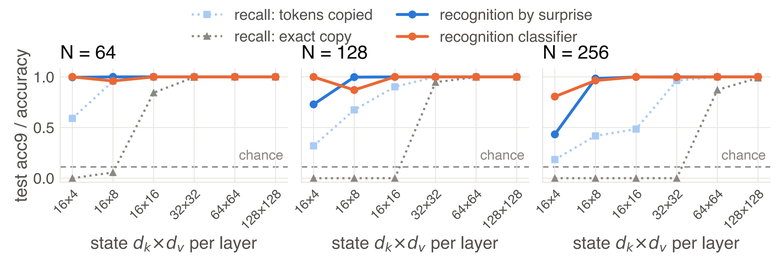


Validation acc9 (full-length sequences) of the 128×128 classifier. Curriculum runs: 3 epochs; no-curriculum runs: 10 epochs (first 11k steps shown).

- **Length curriculum**: acc9 $= 1.000$ at $N=128$ and 256.
- Easy negatives alone fix $N=64$ but **not** $N=256$.
- Comparing learned on short prefixes carries over to long ones.
- In the sweep, **all 42** curriculum-trained classifiers learn; none ends at $\ln 2$.

3 Shrinking the state: recall goes first

6 state sizes (16×4 to 128×128) $\times$ 7 lengths $\times$ 2 objectives: 84 runs, 10 epochs each, tested on 100k sequences.



Test acc9: **recall model's surprise**, **recognition classifier**. Dotted: recall. $N \leq 32$: all ≥0.87.

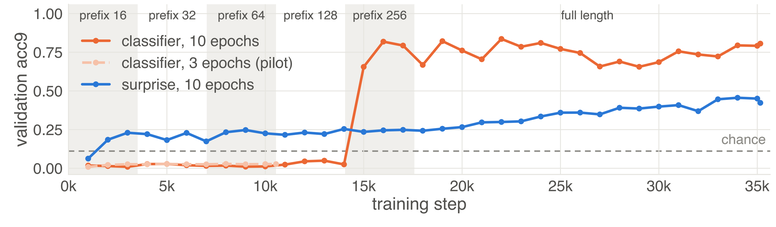
- **Recognition outlasts recall.** At 16×16, $N=256$ the recall model keeps ~half the bits (≥536 of 1,024) and copies no sequence exactly, yet both readouts score acc9 ≥0.999. Surprise reaches 0.98–1.00 while $\geq\frac{1}{3}$ of the bits are kept: above the 8-bit checksum.
- **Smallest state: the classifier wins.**

16×4	recall tokens	surprise acc9	classifier acc9
$N=128$	0.32	0.73	**1.00**
$N=256$	0.18	0.43	**0.81**

Still below a 4-bit checksum (0.85): far from the best code the state could hold.

- At 16×8 surprise is as good or better ($N=128$: 1.00 vs. 0.87).

The 16×4 classifier at $N=256$ learns only once the curriculum reaches full-length prefixes:



Validation acc9, 16×4, $N=256$; shading: curriculum stages. The 3-epoch pilot never left $\ln 2$.

Next

- **Seeds 1–2** at the three smallest states (60 runs): is the 16×4 win real, and the 16×8 dips noise?
- **Longer curriculum** at small states: those runs were still improving.
- **Harder benchmarks** where checksums fail; K stored sequences per context.

Seed 0 only. The three smallest states shrink d_v only (fla kernels need $d_k \geq 16$); across 16×4–16×16 the parameter count changes by 4%.